



CIS 422: Human Computer Interaction
Phase 4

FoodSaver



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Part 1: Heuristic Evaluation using Nielsen's Principles

1.1 Heuristic Evaluation Process

The prototype of FoodSaver was analysed using Nielsen's usability heuristics by another group whose objective was to conduct an analysis of usability of the system in order to identify its weaknesses and strengths. This analysis was performed through the examination of various interfaces in the system as well as examining interactions within the system.

In this analysis, it is important to evaluate whether the usability principles have been adhered to, as well as identify possible problems with the system.

1.2 Heuristic Evaluation Results

1. Visibility of System Status

Example: The list of alerts with time stamps.

strengths: The system keeps the user informed about exactly when things happened. It provides timely feedback about food status.

weaknesses: Some notifications are old but still appear.

Improvement: Add a way to clear or "Mark as read" the notifications to keep the status updated.

2. Match between System and the Real World

Example: The Languages dropdown list.

strengths: It uses a familiar list layout and clear icons.

weaknesses: The list includes countries like "India" as a language. In the real world, "India" is a country, not a language.

Improvement: Update the list to show language names correctly.

3. User Control and Freedom

Example: The Confirmation Pop-up.

strengths: It gives the user a chance to cancel the action. If they clicked "delete" by mistake, they can easily go back without losing their data.

weaknesses: - -

Improvement: - -

4. Consistency and Standards

Example: The Bottom Navigation Bar (Tab Bar).

strengths: It allows users to switch between main sections easily.

weaknesses: There are some spelling mistakes in the labels.

Improvement: Double-check all text and fix the spelling to make the app look more professional.

5. Error Prevention

Example: The Delete account button.

strengths: The button is clearly visible.

weaknesses: Clicking the button deletes the account immediately without a confirmation message.

Improvement: Add a confirmation pop-up or ask the user to type their password before deleting to prevent accidental clicks.

6. Recognition Rather Than Recall

Example:

Expiry date mentioned under each item.

strengths:

The user does not need to memorize any information as all the necessary information is visible on the screen.

weaknesses:

There are no disadvantages associated with this feature.

Improvement:

No improvements are required for this design aspect as it works fine for visibility of information.

7. Flexibility and Efficiency of Use

Example:

Languages menu option within Settings.

Strengths:

It enables users to switch the application language easily according to their preference.

Weaknesses:

It can be hard to navigate through if there are too many options available.

Improvement:

Include a search box within the dropdown menu if the languages list grows very long.

8. Aesthetic and Minimalist Design

Example:

Settings page list.

Strengths:

The list is really simple and easy to understand and use by the user.

Weaknesses:

When you open the dropdown menu from the Settings page, it will overlap with other parts of the website, creating an unclear interface.

Improvement:

Make sure there is enough space between interactive elements.

9. Help Users Recognize, Diagnose, and Recover from Errors

Example:

The Expiration alert banner.

Strengths:

It warns the user regarding the problem and enables him/her to take immediate steps to prevent food wastage.

Weaknesses:

In certain cases, the system does not provide any error message for the user.

Improvement:

Provide simple error messages regarding any problem and tell the user how to solve it.

10. Help and Documentation

Example:

Login procedure and Community tab.

Strengths:

The app employs commonly used language and recognizable icons, making it incredibly easy to comprehend for any individual using it.

Weaknesses:

There is no sign-in process nor a "Get Started" screen to instruct newcomers on creating their profiles.

Improvement:

Create a straightforward Login/Sign-up screen for new users.

1.3 Summary of Findings

The evaluation showed that the FoodSaver prototype is generally easy to use and understandable due to its simple design and familiar layout. However, some usability issues were identified, such as unclear elements, lack of confirmation in important actions, and minor inconsistencies. Improving these areas would enhance the overall user experience.

Part 2: Usability Testing

This section presents the complete usability testing process conducted for the FoodSaver application. Three face-to-face sessions were held with real users representing different target groups identified in Phase 2. The testing covered the full user journey across all major features of the application.

2.1 Overview

The FoodSaver project was made by a team. Each person worked on a different part like the Home screen, Inventory, Calendar, Recipe Suggestions, Community and the Donation and Rewards system. So when we tested the app to see how easy it is to use we wanted to check the thing from start to finish. We did a test on each part of the app to make sure we covered everything.

2.2 Methodology

We used a method called Think-Aloud, where we also watched what people did. We asked people to use a paper version of the app and tell us what they were thinking. The person running the test watched and Did not say anything unless the person got really stuck.

Here are the steps we followed for each test:

- We told the person what the test was for and made sure they knew we were testing the FoodSaver app, not them.
- The person agreed to take the test and let us write down what they said.
- We gave the person one or two things to do using the paper app.
- The person running the test. Wrote down what the person did if they got stuck, made mistakes or said anything.
- After the person finished the tasks we asked them some questions, about what they thought.
- At the end we talked a bit about what they thought of the FoodSaver app overall.

2.3. Test Setup

This section outlines the setup and environment used for the usability testing sessions. It provides information on the testing conditions, tools employed, participant engagement, and data collection methods. These details were carefully structured to maintain an organized and consistent approach throughout all sessions.

Aspect	Description
Environment	Usability testing sessions took place in a calm, comfortable setting to reduce distractions and help participants concentrate fully on their tasks.
Prototype Type	A paper prototype for the FoodSaver application was utilized.
Device Simulation	The prototype was presented with a smartphone-style layout to simulate real mobile interaction.
Interaction Method	Participants interacted directly with the prototype while they carried out their assigned tasks.
Testing Method	The Think-Aloud method was employed, where participants verbalized their thoughts while performing the tasks.
Facilitator Role	The facilitator guided the participant through the tasks, provided instructions, and observed without interfering unless the participant needed help.
Observation Method	Participants were observed directly during the tasks, focusing on their behavior, difficulties, and reactions.
Data Collection	Data was collected by taking notes on user actions, errors, and verbal feedback during the session.
Recording	No audio or video recording was used; only written notes were taken during the sessions.
Session Duration	Each session lasted approximately 10–15 minutes.

The table above summarizes the usability testing setup.

2.4 Participants

Three participants were recruited to represent different segments of FoodSaver's target user groups. Participant selection was guided by the user profiles identified in the Phase 2 user analysis.

#	Name	Profile	Age Group	Relevance to FoodSaver
1		University Student	18–24	Represents young adults who use food apps for convenience and sustainability
2		University Student	18–24	Represents users who manage personal food inventory and meal planning
3		Housewife & Mother	35–44	Represents household food managers — the primary target user of FoodSaver

2.5 Tasks

Each participant was assigned tasks relevant to the area of the app being tested. Tasks were designed to be realistic, short, and representative of common usage scenarios identified in Phase 2.

#	Participant	Task Description	Area Tested
1		Navigate to the Donation section, complete a food donation to a local charity, and review the rewards received after donating.	<i>Home Screen — Donate Food Sub-screens & Rewards System</i>
2		Review the food inventory on the Home screen, identify items that are expiring soon, and use the calendar to schedule a future date.	<i>Home Screen — Inventory & Calendar</i>
3		Open the app, select available ingredients from the inventory, and find a suitable recipe based on those ingredients.	<i>Inventory Screen & Recipe Suggestions</i>

University Student | Donation & Rewards

Participant Profile

Name	
Background	University student with regular smartphone usage
Prior App Experience	No prior experience with FoodSaver
Reason for Selection	Represents the 18–24 age group — 42.9% of target users in Phase 2 survey

Task

Navigate to the Donation section, complete a food donation to a local charity, and review the rewards received after donating.

Session Observations

completed the donation task in less than 20 seconds from the Home screen to the confirmation screen. She navigated directly to the Donate Food Quick Action button without any hesitation, selected the charity Eitdal (Saudi Food Bank), chose items from the list, and confirmed the donation. Her behavior indicated that the navigation flow between screens is intuitive and the labels are easy to understand.

Upon reaching the Donation Sent confirmation screen, paused and read the screen carefully. She liked the badge rewards that said "Hero" and "Supporter" they made donating feel fun also she said the screen changes were smooth.

The important buttons were easy to find.

However, during the debrief, mentioned that she completed the donation very quickly without reviewing the selected items carefully. She only saw the Donation Summary on the confirmation screen — after the donation was already submitted — and realized she had not checked which items were being donated beforehand.

Findings & Analysis

Finding 1 — Reward Badges Successfully Motivate Donation Behavior [STRENGTH]

Observation: We noticed that _____ really liked the badges she gets on the donation sent screen. She mentioned that these small rewards made the whole process feel more like a fun achievement and not just a boring task.

Analysis: This part really works because it follows the Fogg Behavior Model we already studied in Phase 2. When the app gives a reward right after the user finishes a task, it encourages them to do it more. Seeing badges like hero or supporter makes the user feel very good about their donation, so they will probably want to come back and use the app again. So we should keep the reward system. Think about adding more rewards for other.

Recommendation: We definitely need to keep these badges. It is also a good idea to add more of them for different things, like when someone is really good at updating their inventory or if they cook something using food that's almost expired.

Finding 2 — No Pre-Submission Review Step Before Confirming Donation [ISSUE]

Observation: During the testing, we noticed that _____ finished the entire donation process without a chance to double-check her choices. She only saw the summary of what she donated after the process was already completed and sent.

Analysis: Since there is no review step right now, users might easily select the wrong item or charity without noticing. If they click confirm by mistake, they can not go back and change it. Adding a way to double check will help prevent these kinds of accidents.

Recommendation: We can just add a simple pop up or a small screen that shows a summary before the final step to review everything one last time. It should list what they picked like the items, how many, and the charity—so users like _____ can be 100% sure before they send the donation.

Session Result

Task Completion	Completed successfully
Time on Task	Under 20 seconds
Errors Encountered	None during task execution
Satisfaction	High positive verbal feedback
Key Strength	Reward badges are motivating
Key Issue	No pre-submission review step

Test Session 2 —

University Student | Inventory & Calendar

Participant Profile

Name	
Background	University student who manages her own food at home
Prior App Experience	No prior experience with FoodSaver
Reason for Selection	Represents users who manage personal food inventory identified in Phase 2 as a key user group

Task

Review the food inventory on the Home screen, identify items that are expiring soon, and use the calendar to schedule a future date.

Session Observations

was able to navigate through the application and complete the assigned task. However, she encountered two noticeable usability issues during the session that required additional time and caused visible confusion.

On the Home screen, did not immediately notice the expiry warning for food items. She scrolled past the warning area without pausing. When the facilitator asked her to look for expiring items, she returned to the Home screen and eventually located the warning, but admitted she had not noticed it initially. When asked why, she explained that the grey color of the expiration tag made it look like inactive or unimportant information.

On the Calendar screen, successfully located the highlighted expiry date marked with a red border. However, she hesitated when she saw the past dates and tried tapping on one of them. She was unsure whether past dates were interactive or not, and expressed confusion about what they would do if tapped. She suggested that past dates should be visually different to make it clear they are not selectable.

Findings & Analysis

Finding 1 — Expiry Warning Color Is Too Subtle [ISSUE]

Observation: did not notice the expiry warning on the Home screen during her first pass. She explained that the grey color made it appear unimportant or disabled.

Analysis: Color is one of the most immediate visual cues for communicating urgency and system status. Grey is conventionally associated with inactive or non-interactive elements, which caused to overlook the alert entirely. This is a clear violation of Nielsen's heuristic of 'Visibility of System Status' the system failed to communicate an important warning effectively. It is also connected to the survey finding in Phase 2 that 48.6% of users throw away food without noticing expiry, suggesting this issue could have real world consequences.

Recommendation : Replace the grey expiry tag color with a high contrast warning color such as red or orange. Add a warning icon for example triangle with new color and exclamation mark to make the alert immediately noticeable without relying on icon alone.

Finding 2 — Past Calendar Dates Are Visually Ambiguous [ISSUE]

Observation: was confused about whether past calendar dates were interactive. She attempted to tap them without knowing what would happen, and suggested they should be faded out.

Analysis: The lack of visual differentiation between past and future calendar dates creates ambiguity that violates the heuristic of 'Match Between System and the Real World.' In real-world calendars and most digital calendar apps, past dates are greyed out or visually de-emphasized to signal they are not actionable. Additionally, this violates the heuristic of 'Consistency and Standards' users expect past dates to be non-interactive based on their prior experience with other apps.

Recommendation: Apply a faded or light grey style to all past calendar dates to clearly communicate that they are not selectable. This change is low-effort but would significantly reduce confusion for all users.

Participant Quote:

"I want the colors to help me . I want things that are important to be bright and things that are not important to be faded out."

Session Result

Task Completion	Completed with noticeable hesitation
Time on Task	Longer than expected due to two usability issues
Errors Encountered	Failed to notice expiry warning on first pass; attempted to tap non-interactive past dates
Satisfaction	Moderate completed the task but expressed frustration with color usage
Key Issue 1	Expiry warning color is too subtle (grey)
Key Issue 2	Past calendar dates are visually ambiguous

Test Session 3 —

Housewife & Mother | Inventory & Recipe Suggestions

Participant Profile

Name	
Background	Housewife and mother responsible for daily household food management and cooking
Prior App Experience	No prior experience with FoodSaver
Reason for Selection	Represents the primary target user of FoodSaver — Phase 2 identified parents (42.9%) as the largest user group. Her daily food management habits make her feedback especially valuable.

Task

Open the app, select available ingredients from the inventory, and find suitable recipe based on those ingredients.

Session Observations

went to the Inventory screen by herself. Right away understood how to use the checkboxes to select things. She picked the ingredients she had in the Vegetables, Fruits and Dairy categories without any help. Then moved to the Recipe Suggestions screen.

On the Recipe Suggestions screen really liked the ingredient tags at the top of the screen. She thought this was an idea because she did not have to go back and check what ingredients she had chosen. The little badges that showed how well the ingredients matched on each recipe card also caught her eye. She said some nice things about them.

clicked on the Spinach Omelette recipe. Looked at the details. She read what other people were saying about the recipe. Said that it made her trust the recipe more because she could see what other users thought.

When we talked about it afterwards had an idea for how to make the app better. She said it would be really helpful if the app could show her which ingredients she was missing for each recipe. That way she could add them to her list when she went shopping time. She said that as a housewife she likes to plan meals of time and knowing what to buy would make the app something she could use every day with the Inventory screen and the Recipe Suggestions screen and the recipe detail screen. thought the app was good. It would be even better if it could help her with her shopping list, on the Inventory screen.

Findings & Analysis

Finding 1 — Ingredient Tags Effectively Reduce Memory Load [STRENGTH]

Observation: immediately noticed and appreciated the ingredient tags at the top of the Recipe Suggestions screen, which showed what she had selected in the Inventory. She said it saved her from having to go back and check.

Analysis: This finding validates the design decision to show ingredient tags on the Recipe screen. It directly addresses the UCD principle of reducing short-term memory load identified in Phase 2, where users should not be required to remember information from one screen to the next. The fact that a housewife who handles things at once finds this feature really helpful shows it's valuable in real life.

Recommendation: Maintain the ingredient tags also should stay visible this helps when users scroll down the recipe list the ingredient tags must remain on screen they are important for users to see.

Finding 2 — Missing Ingredients Are Not Shown [ISSUE]

Observation: During the testing session, noticed that while the match percentage was clear the app did not show what was actually missing for the recipe. She mentioned that for a housewife who needs to plan her grocery shopping in advance, this information is very important to know exactly what she needs to buy.

Analysis : If we look at the current design, the app shows a match percentage 95%, but it does not tell the user what is actually missing. This makes it a bit hard for users who want to plan their meals properly. Fixing this helps our main goal in FoodSaver which is to stop food waste by making sure people only buy the ingredients they really need.

Recommendation: It would be much better to add a small note like you are missing some items under the percentage on the recipe card. This way it is easier for anyone to check what they need and put it on their shopping list immediately



Session Result

Task Completion	Completed successfully and confidently
Time on Task	Within expected range no hesitation
Errors Encountered	None
Satisfaction	High positive feedback throughout the session
Key Strength	Ingredient tags reduce memory load effectively
Key Issue	Missing ingredients are not shown limits meal planning

Overall Summary of Findings

The three usability testing sessions revealed a total of six findings three strengths and three issues. All three participants were able to complete their assigned tasks, indicating that the overall navigation structure of FoodSaver is sound. The issues identified are specific and addressable, and will be resolved in the Final Design Update section.

#	Participant	Task	Result	Strength	Issue
1	(Student)	Donation & Rewards	Success	Reward badges motivate donation behavior	No pre-submission review step
2	(Student)	Inventory & Calendar	Success	Red highlight on expiry dates is clear	Expiry warning color too subtle; past dates ambiguous
3	(Housewife)	Inventory & Recipes	Success	Ingredient tags reduce memory load	Missing ingredients are not shown

Part 3: Final Design Update

Settings Interface:

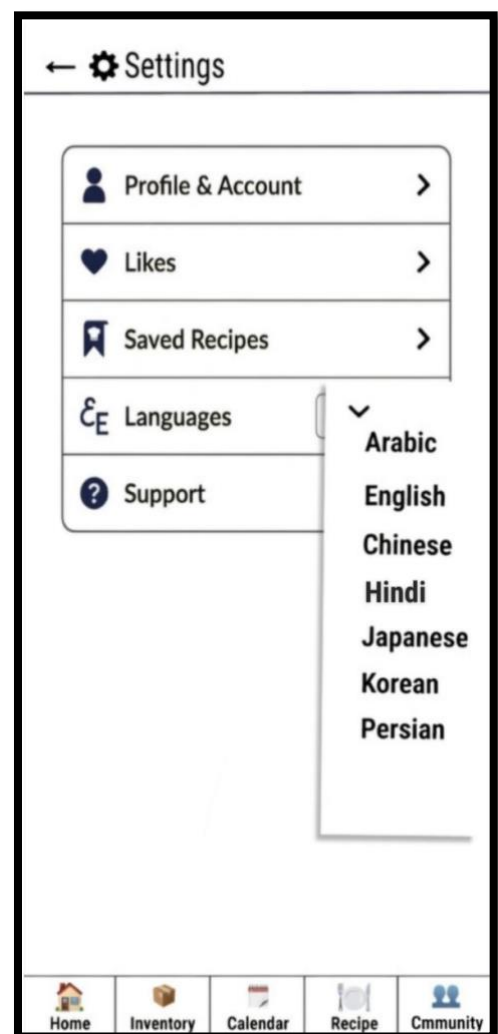
weaknesses: The list includes countries like "India" as a language. In the real world, "India" is a country, not a language.

Improvement: Update the list to show language names correctly.

Before:



After:



Before: There was a mistake in one of the language names in the menu.

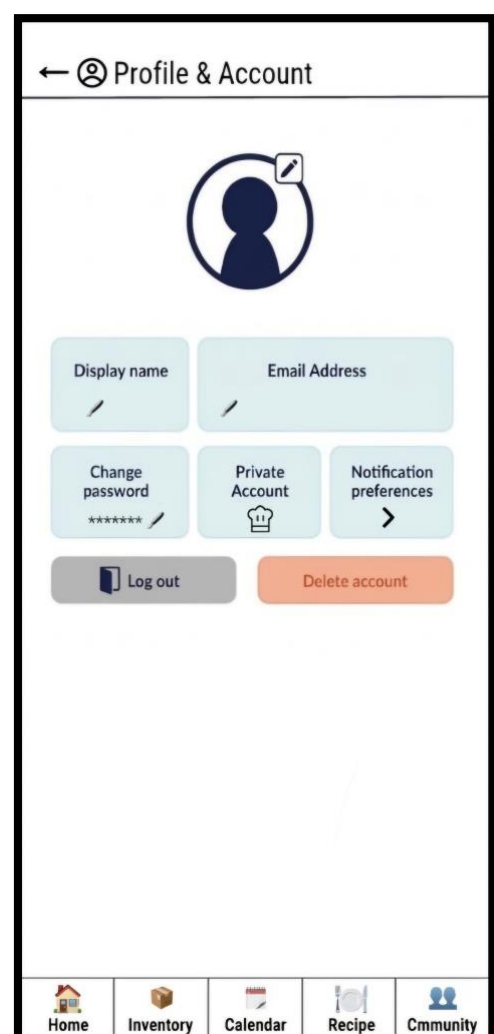
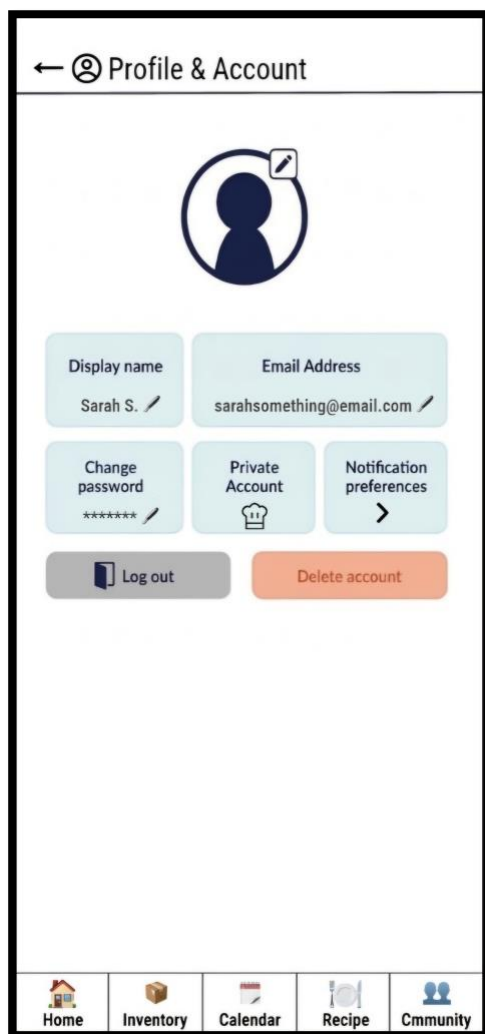
After: I update the menu to show the correct language name.

Profile & Account Interface:

weaknesses: Clicking the button deletes the account immediately without a confirmation message.

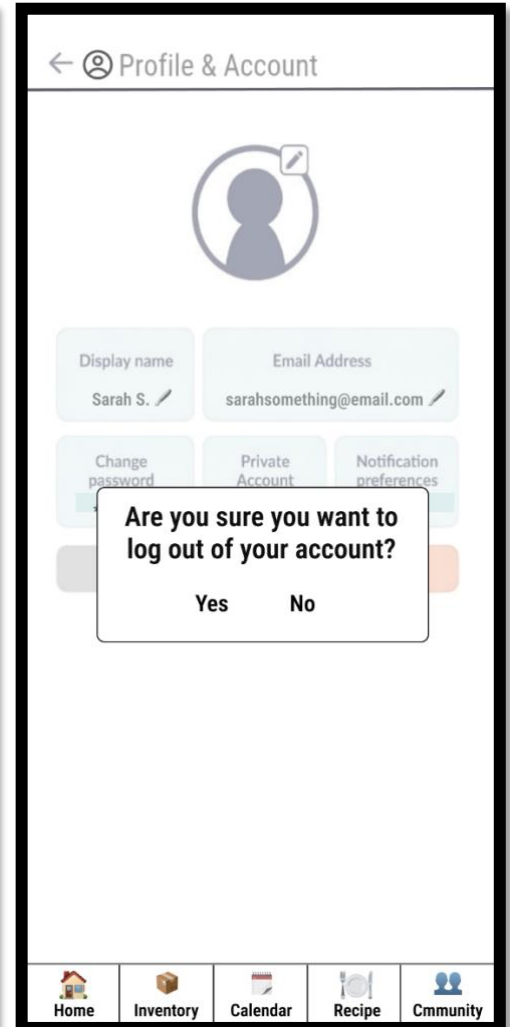
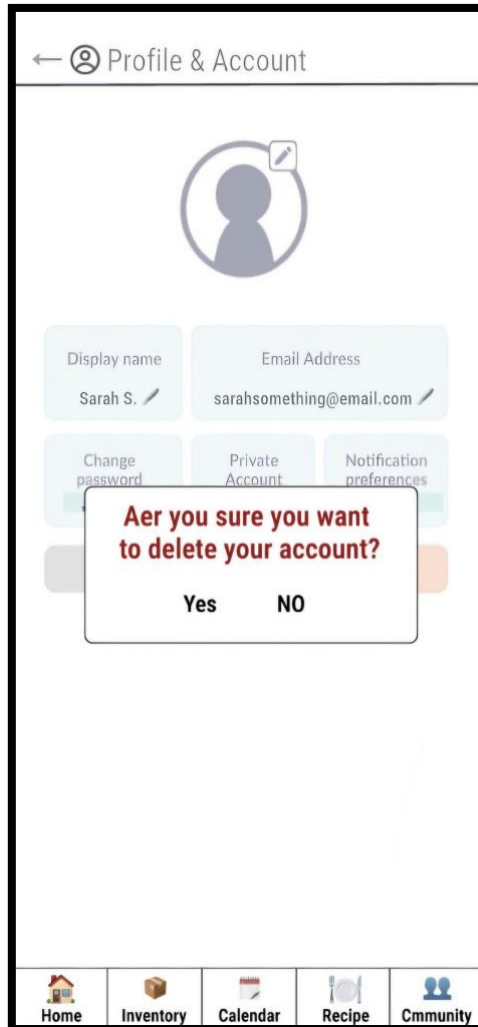
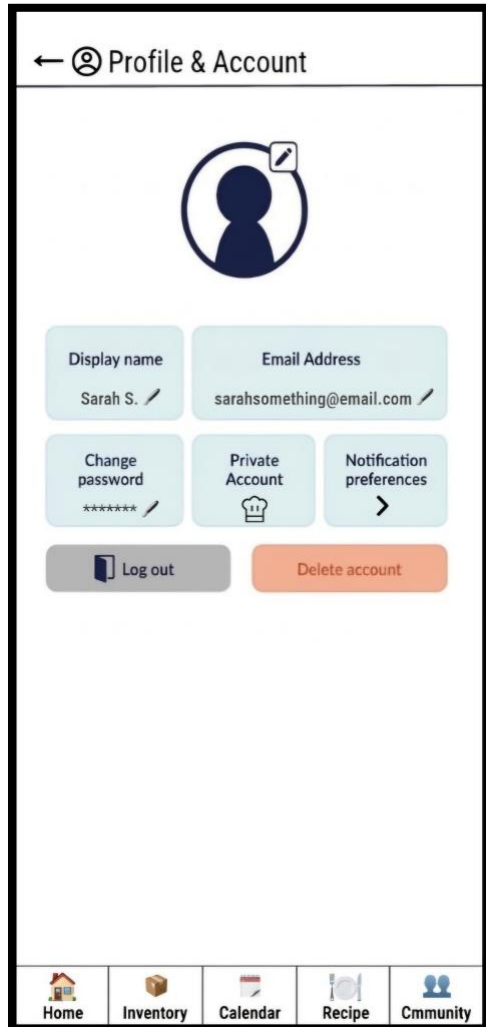
Improvement: Add a confirmation pop-up or ask the user to type their password before deleting to prevent accidental clicks.

Before:



Before: There was no confirmation pop up for account deletion; it would simply be deleted.

After:



After: Now, I've added a confirmation pop up before deletion or log out; only if the user clicks “Yes” will the actions proceed.

Profile & Account Interface:

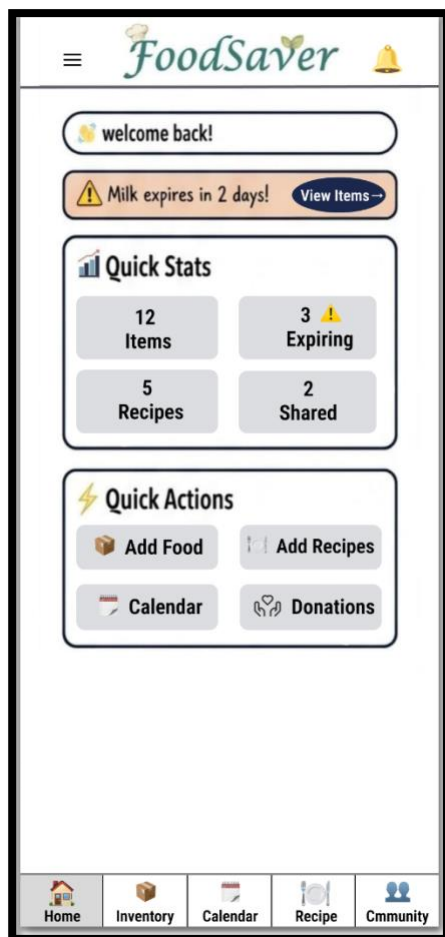
Weaknesses:

There is no sign-in process nor a “Get Started” screen to instruct newcomers on creating their profiles

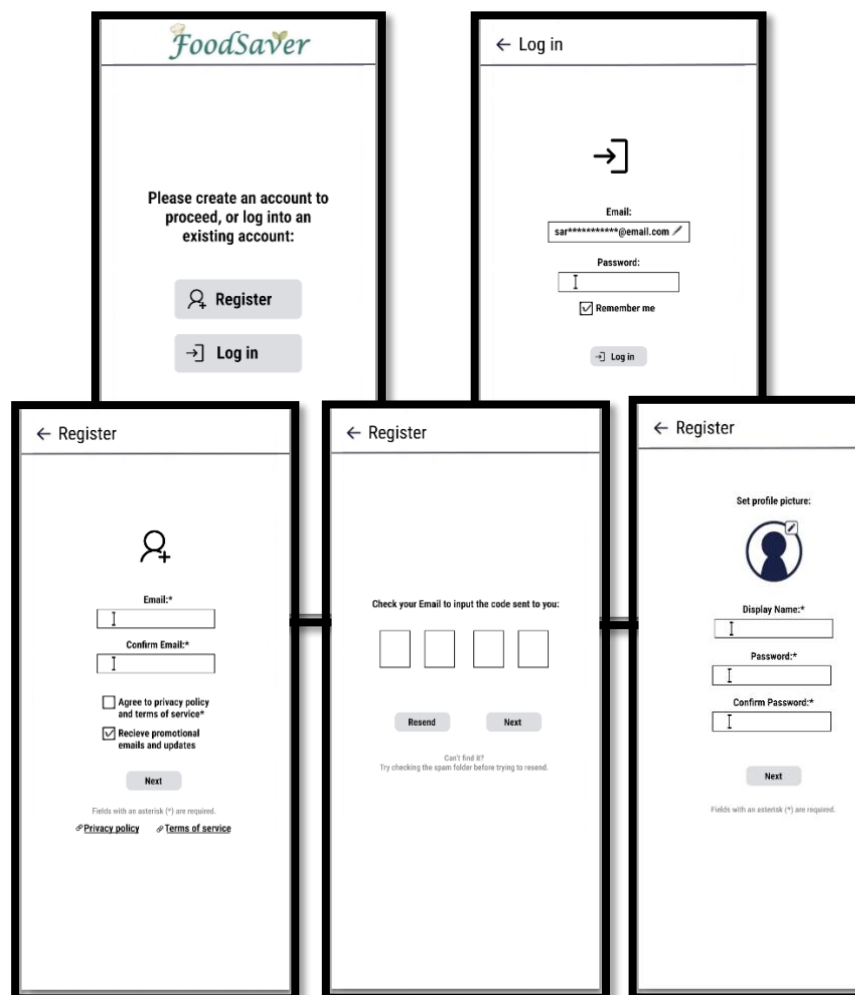
Improvement:

Create a straightforward Login/Sign-up screen for new users.

Before:



After:



Before: There was no process implemented, nor a way to create a new account.

After: Added standardized login and sign-up step by step screens so it's clear to non tech savvy users and familiar to most users, and accessible by default after logging out or deleting your account.

Settings Interface:

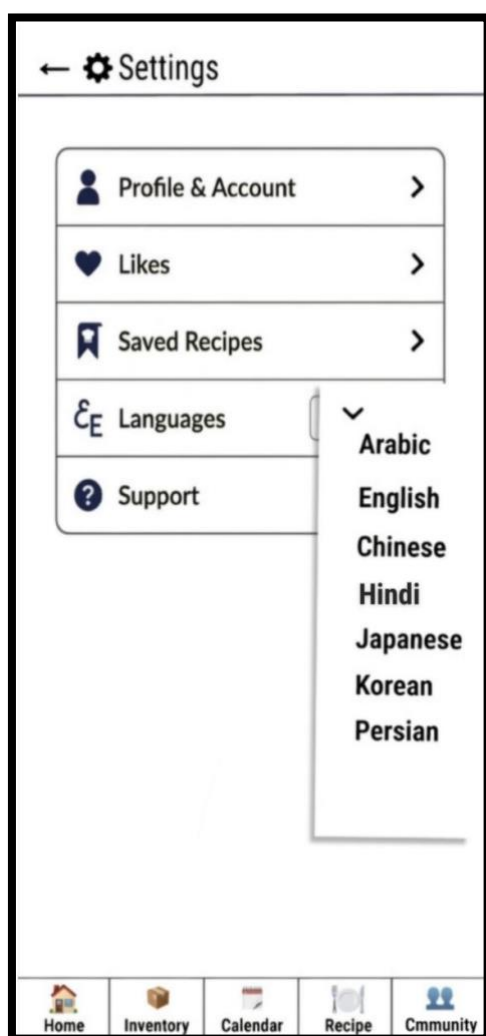
Weaknesses:

It can be hard to navigate through if there are too many options available.

Improvement:

Include a search box within the dropdown menu if the languages list grows very long.

Before:



After:



Before: The language list was long and difficult to search in it.

After: Now I add a search box inside the menu so the user can search for the language instead of scrolling through the list.

Home Interface:

Weaknesses:

When you open the dropdown menu from the Settings page, it will overlap with other parts of the website, creating an unclear interface.

Improvement:

Make sure there is enough space between interactive elements.

Before:



After:



Before: There was a layering problem, when the user open the menu the header line appeared in top of it creating in clear look.

After: I fixed the layers so the line no longer overlaps with the menu.

Expiry Calendar Interface:

Weaknesse:

In certain cases, the system does not provide any error message for the user.

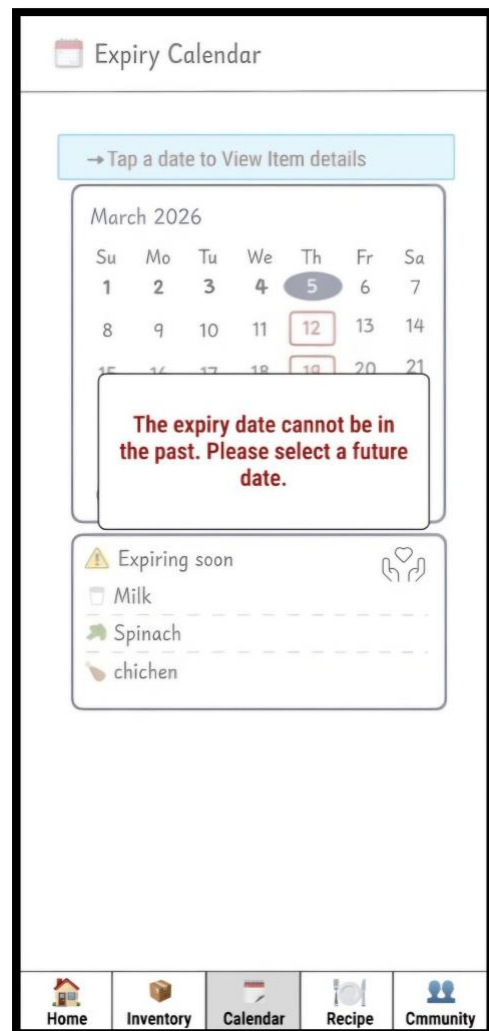
Improvement:

Provide simple error messages regarding any problem and tell the user how to solve it.

Before:



After:



Before: The past dates were able to selected like other dates, with no error message explain why they Didn't work.

After: now when the user clicks a past dates, clear error message appears for five seconds telling Them how to fixed the issue.

Expiry Calendar Interface:

Weakness:

Past Calendar Dates Are Visually Ambiguous.

Improvement:

Apply a faded or light grey style to all past calendar dates to clearly communicate that they are not selectable .

Before:



After:



Before: The past dates were black and looked like other dates, which was confusing.

After: I changed past dates to grey for the user knew that they cannot selected.

Recipe Interface:

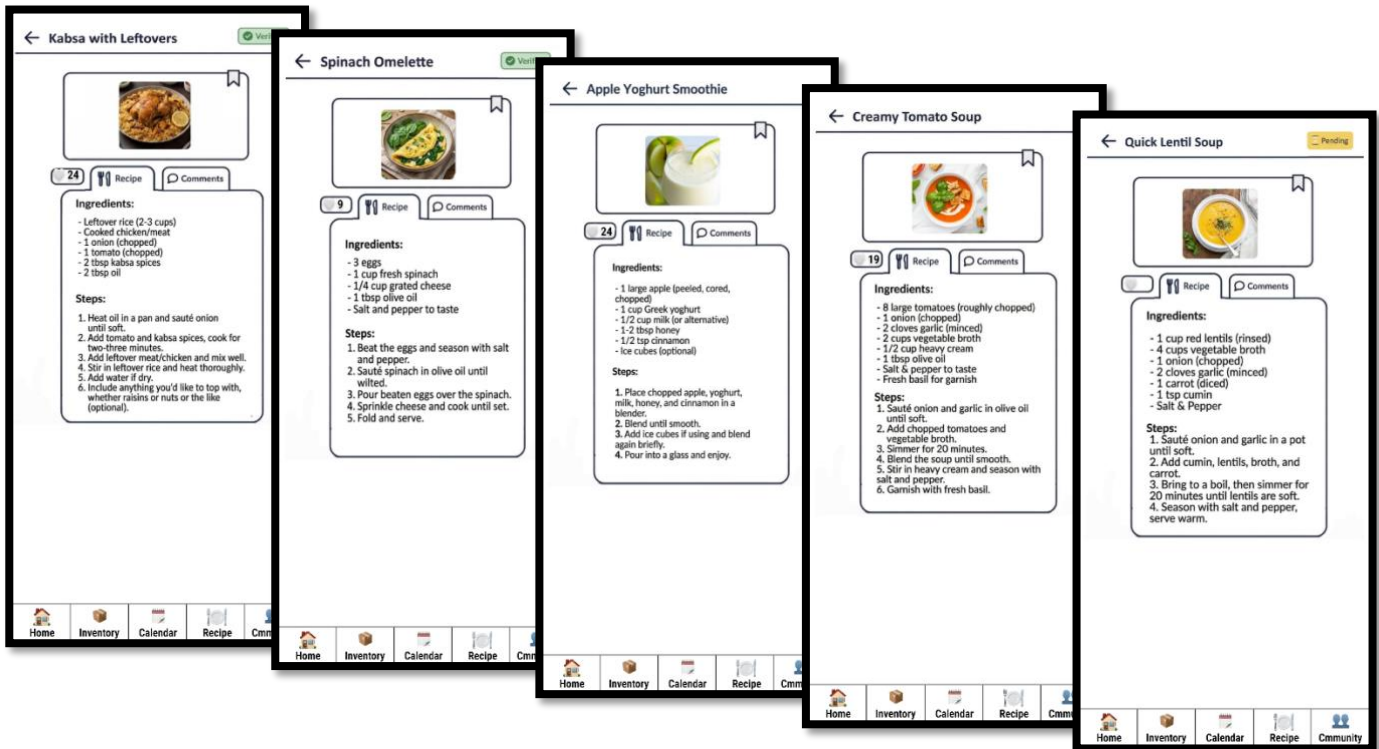
Weakness:

There is no indication of missing recipes.

Improvement:

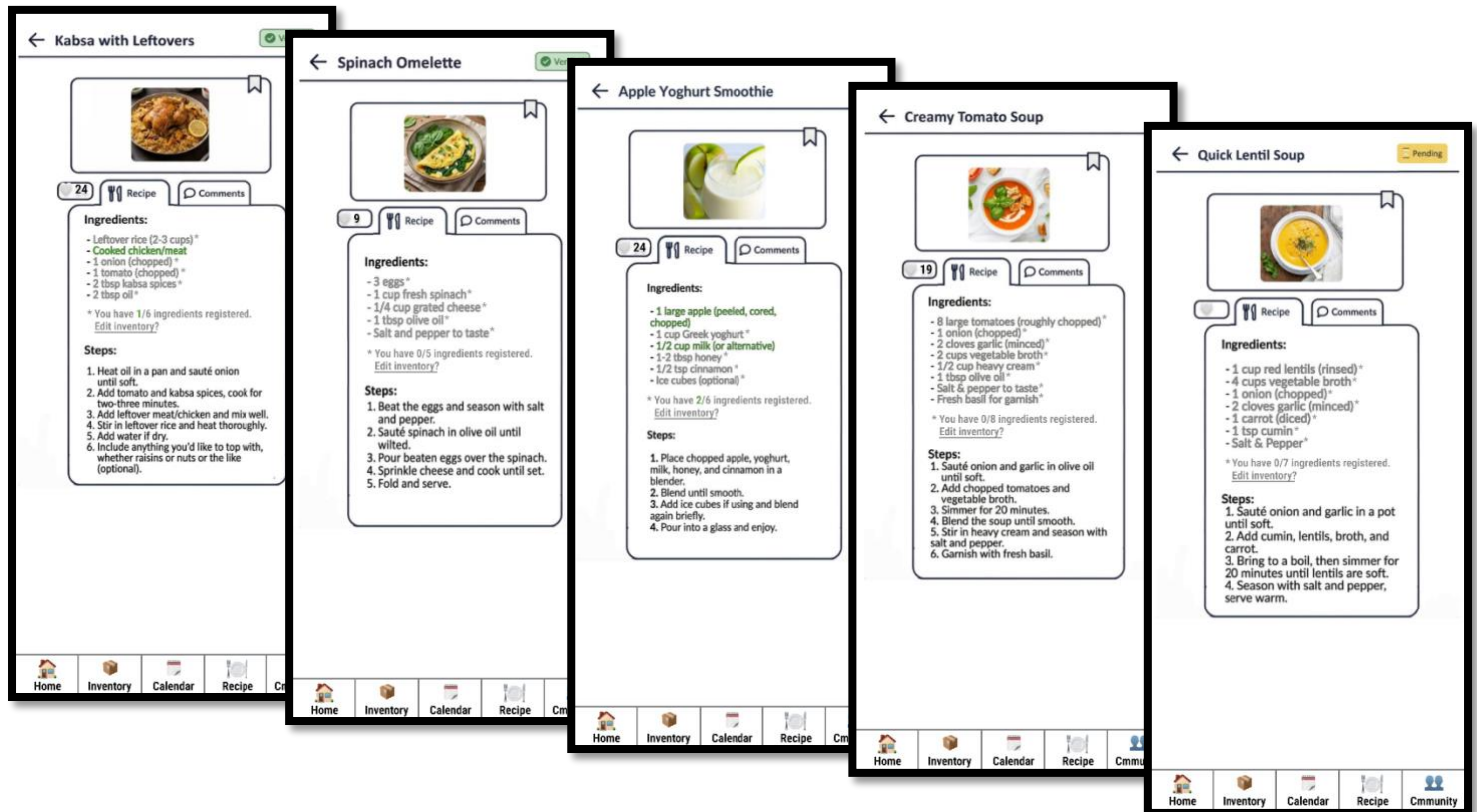
Apply a notable difference between registered items and unregistered items for extra clarity to the users considering the recipes.

Before:



Before: No visible indicator that references the user's inventory.

After:



After: Changed the colors of registered/unregistered food items, and gave a shortcut to edit the list.

Notification interface:

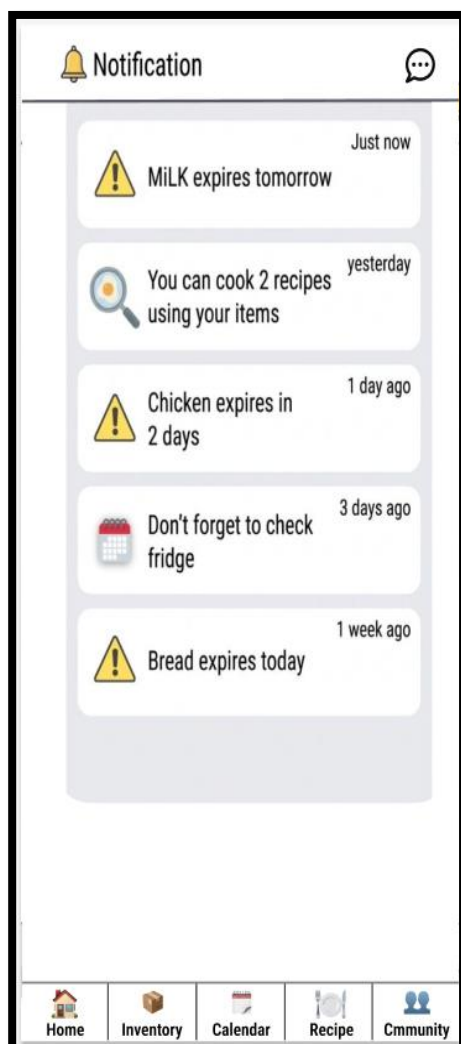
Weaknesses:

Some notifications are old but still appear.

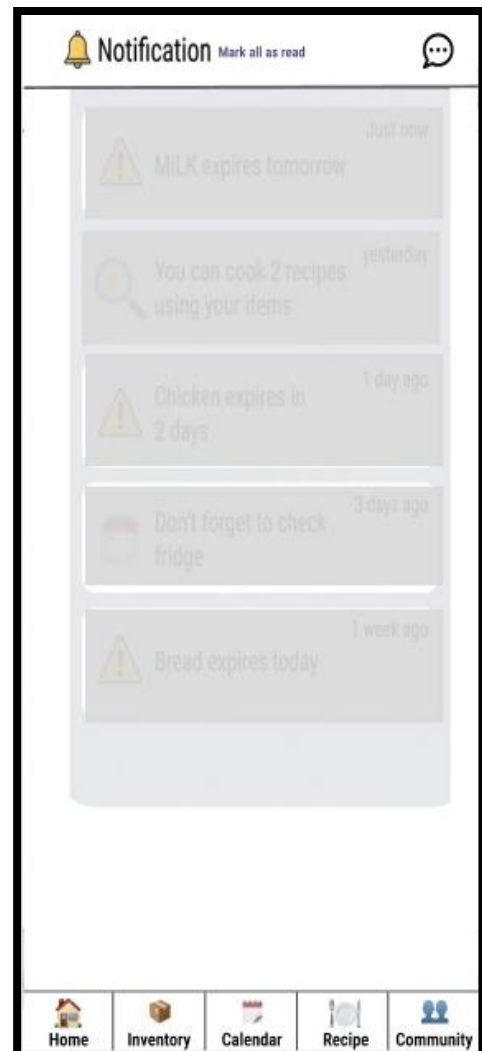
Improvement:

Add a way to clear or “Mark as read” the notification to keep the status update.

Before:



After:



Before: There was no mark all as read and the notification were showing and not going

After: There is a mark all as read button, and when you click on it, notification start going

Dictation correction:

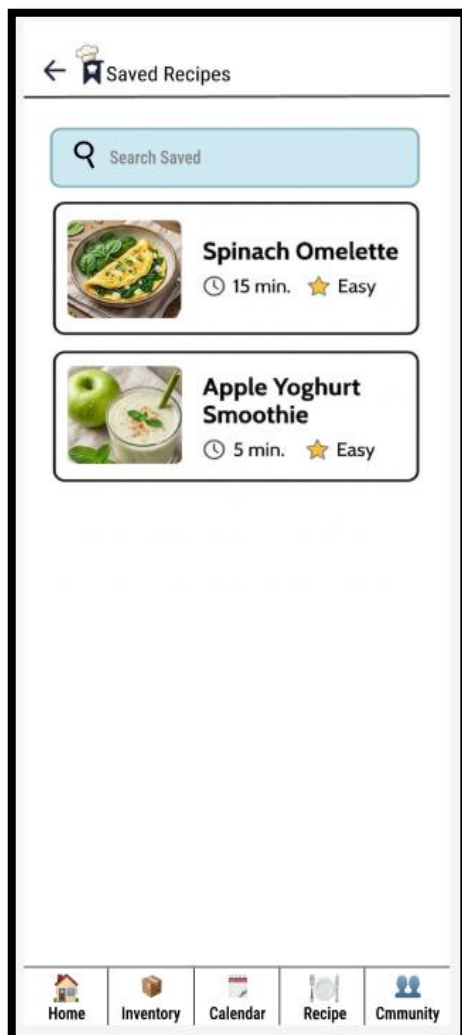
Weaknesses:

There are some spelling mistakes in the labels.

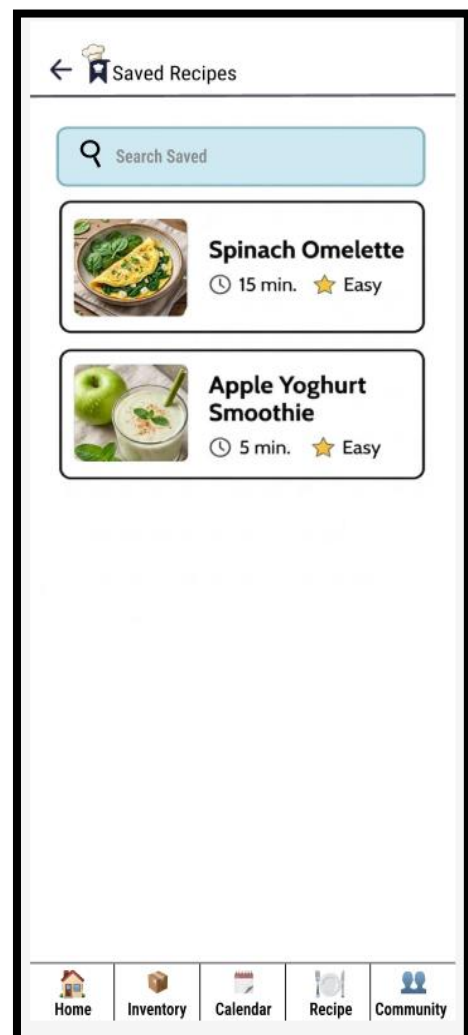
Improvement:

Double-check all text and fix the spelling to make the app look more professional.

Before:



After:



Before: There was a spelling mistake in the word community.

After: The word has been modified in all pages with modification of the format after correcting the word to become more professional.

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prototype link:

<https://www.figma.com/design/Xtg1HQsOoXyGRVRG9GWx7d/Untitled?node-id=0-1&t=bxg6QSPFaWS7hQxy-1>